
MAT 527: Functions of a Complex Variable I

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Homework V

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Problem 1 Compute the improper integral

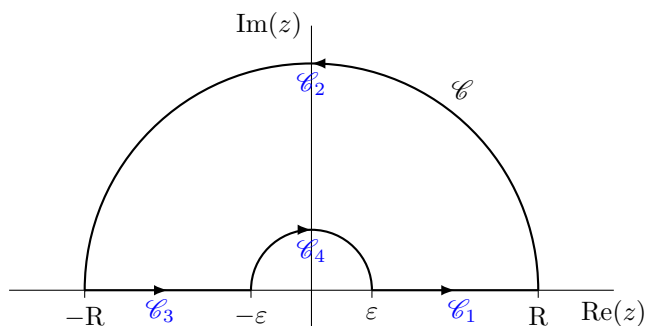
$$\int_0^{\infty} \frac{\sin x}{x} dx$$

(a) using Cauchy's Theorem and (b) using Cauchy's Integral Formula.

Solution First note that $\sin(x) = \operatorname{Im}(e^{ix})$. Thus,

$$\int_0^{\infty} \frac{\sin x}{x} dx = \operatorname{Im} \left(\int_0^{\infty} \frac{e^{ix}}{x} dx \right). \quad (1)$$

To compute the integral on the right-hand side of (1), first let $f(z) = \frac{e^{iz}}{z}$, which is holomorphic everywhere except at $z = 0$ (since it is a ratio of entire functions). Next, define $\mathcal{C}$ as the closed, simple contour given in the figure below:



Since $\mathcal{C}$ does not contain $z = 0$, our function $f(z)$ is holomorphic in and on $\mathcal{C}$. Therefore

$$\int_{\mathcal{C}} f(z) dz = 0, \quad (2)$$

by Cauchy's Theorem. The integral in (2), however, can be considered as the sum of four integrals along the four sections of the contour $\mathcal{C}$ (i.e., $\mathcal{C}_1$, $\mathcal{C}_2$, $\mathcal{C}_3$, and $\mathcal{C}_4$). The integrals along $\mathcal{C}_1$ and $\mathcal{C}_2$ will become real integrals, while the integrals along $\mathcal{C}_2$ and $\mathcal{C}_4$ will need to be parameterized. The contour $\mathcal{C}_2$ can be parameterized by letting $z = Re^{it}$ and letting t range from 0 to π , whereas $\mathcal{C}_4$ can be parameterized by letting $z = \epsilon e^{it}$ and letting t range from π to 0. Hence,

$$\int_{\mathcal{C}} f(z) dz = \int_{\mathcal{C}_1} f(z) dz + \int_{\mathcal{C}_2} f(z) dz + \int_{\mathcal{C}_3} f(z) dz + \int_{\mathcal{C}_4} f(z) dz \quad (3)$$

$$= \int_{\varepsilon}^R \frac{e^{ix}}{x} dx + \int_0^{\pi} f(Re^{it}) Rie^{it} dt + \int_{-R}^{-\varepsilon} \frac{e^{ix}}{x} dx + \int_{\pi}^0 f(\varepsilon e^{it}) \varepsilon ie^{it} dt \quad (4)$$

$$= \int_{\varepsilon}^R \frac{e^{ix}}{x} dx + \int_{\varepsilon}^R \frac{e^{-ix}}{-x} dx + \int_0^{\pi} \frac{e^{iRe^{it}}}{Re^{it}} Rie^{it} dt + \int_{\pi}^0 \frac{e^{i\varepsilon e^{it}}}{\varepsilon e^{it}} \varepsilon ie^{it} dt \quad (5)$$

$$= \int_{\varepsilon}^R \frac{e^{ix} - e^{-ix}}{x} dx + i \int_0^{\pi} e^{iRe^{it}} dt - i \int_0^{\pi} e^{i\varepsilon e^{it}} dt \quad (6)$$

$$= 0. \quad (7)$$

It can be shown that the second integral in (6) converges to 0 as R tends to infinity. Note that

$$\left| \int_0^{\pi} e^{iRe^{it}} dt \right| = \left| \int_0^{\pi} e^{iR \cos t} e^{-R \sin t} dt \right| \quad (8)$$

$$\leq \int_0^{\pi} |e^{iR \cos t}| |e^{-R \sin t}| dt \quad (9)$$

$$= \int_0^{\pi} e^{-R \sin t} dt \quad (10)$$

$$= 2 \int_0^{\frac{\pi}{2}} e^{-R \sin t} dt. \quad (11)$$

Now consider the function $g(t) = \frac{2R}{\pi}t$. Note that $g(0) = 0 = R \sin(0)$ and $g(\frac{\pi}{2}) = R = R \sin(\frac{\pi}{2})$. This, along with the fact that $\sin t$ is continuous and concave down on the interval $(0, \frac{\pi}{2}]$, means that $R \sin t \geq \frac{2R}{\pi}t$ on $[0, \frac{\pi}{2}]$. Thus, $e^{-\frac{2R}{\pi}t} \geq e^{-R \sin t}$ on $[0, \frac{\pi}{2}]$. Therefore,

$$2 \int_0^{\frac{\pi}{2}} e^{-R \sin t} dt \leq 2 \int_0^{\frac{\pi}{2}} e^{-\frac{2R}{\pi}t} dt \quad (12)$$

$$= \left[\frac{-2\pi}{2R} e^{-\frac{2R}{\pi}t} \right]_0^{\frac{\pi}{2}} \quad (13)$$

$$= \frac{\pi}{R} [1 - e^{-R}], \quad (14)$$

which tends to 0 as R tends to infinity. Thus, it must also be true that $\int_0^{\pi} e^{iRe^{it}} dt \rightarrow 0$ as $R \rightarrow \infty$. This means that (6) can now be rewritten as

$$\int_{\varepsilon}^{\infty} \frac{e^{ix} - e^{-ix}}{x} dx = i \int_0^{\pi} e^{i\varepsilon e^{it}} dt. \quad (15)$$

We can rewrite the right-hand side of (15) using a Maclaurin series. That is

$$e^{i\varepsilon e^{it}} = 1 + i\varepsilon e^{it} + \mathcal{O}(i\varepsilon e^{it}) i\varepsilon e^{it},$$

where $\mathcal{E}(w)$ tends to 0 as w tends to 0. Therefore,

$$\int_{\varepsilon}^{\infty} \frac{e^{ix} - e^{-ix}}{x} dx = i \int_0^{\pi} 1 dt - \int_0^{\pi} \varepsilon e^{it} dt - \int_0^{\pi} \mathcal{E}(i\varepsilon e^{it}) \varepsilon e^{it} dt \quad (16)$$

$$= i\pi - \int_0^{\pi} \varepsilon e^{it} dt - \int_0^{\pi} \mathcal{E}(i\varepsilon e^{it}) \varepsilon e^{it} dt. \quad (17)$$

We now show that the two integrals in (17) tend to 0 as $\varepsilon \rightarrow 0^+$. First, let $a > 0$. Choose $\delta = \frac{a}{\pi}$. Then whenever $0 < \varepsilon < \frac{a}{\pi}$, we have

$$\left| \int_0^{\pi} \varepsilon e^{it} dt - 0 \right| \leq \int_0^{\pi} |\varepsilon e^{it}| dt \quad (18)$$

$$= \int_0^{\pi} \varepsilon dt \quad (19)$$

$$= \varepsilon \pi \quad (20)$$

$$< \frac{a}{\pi} \pi \quad (21)$$

$$= a. \quad (22)$$

Therefore, $\int_0^{\pi} \varepsilon e^{it} dt \rightarrow 0$ as $\varepsilon \rightarrow 0^+$.